

MULTIVARIABLE OPTIMIZATION GROUP PROJECT

Project Overview:

In this group project, you will tackle one of several real-world case studies (See the list below and pick one case for your project) that require the application of multivariable functions, optimization, and constrained optimization techniques. The goal is to synthesize theoretical knowledge with practical applications, allowing you to experience the direct impact of mathematical optimization in various industries. This assignment will be 15% of the total grade of the course.

Case Studies Options:

1. **Manufacturing:** Optimize a product mix to maximize profitability under resource constraints.
2. **Machine Learning:** Adjust neural network weights to optimize performance on a given dataset.
3. **Environmental Engineering:** Optimize wastewater treatment processes to balance efficiency with environmental regulations.
4. **Logistics:** Optimize delivery routes for efficiency under time and distance constraints.
5. **Energy Management:** Develop a model to optimize energy consumption in a smart grid system, taking into account fluctuating energy prices, demand cycles, and the incorporation of renewable energy sources.
6. **Healthcare Logistics:** Create a model to optimize the allocation of medical resources, such as staff, equipment, and beds, in a hospital to improve patient throughput while maintaining a high level of care under budgetary and spatial constraints.

Project Deliverables:

Report: A comprehensive report detailing your problem-solving approach, mathematical models used, solution methodology, potential real-world applications, and reflections on the limitations and implications of your findings (10-15 pages).

Video Presentation: A 2–3-minute video summarizing your project's objectives, methods, results, and significance.

Theoretical Component

- Provide a literature review of optimization methods relevant to your case study.
- Develop a mathematical model of your problem, using multivariable functions and demonstrating the use of optimization and constrained optimization.

Practical Application:

- Apply your theoretical model to a practical scenario provided by the project brief.
- Use simulations, algorithms, or real-world data to test and refine your model.
- Discuss the practicality and scalability of your solution in the industry context.

Ethical Use of ChatGPT and Generative AI:

As part of this project, you may or may not utilize generative AI tools. Adhere to the following ethical guidelines:

Attribution: Clearly credit any AI-generated content or assistance in your project.

Originality: Use AI as a tool for ideation and problem-solving, not to replace your original thought process and contribution.

Integrity: Do not use AI to generate misleading or falsified content or data.

Fair Use: Use AI-generated outputs within the scope of fair use, avoiding copyright infringements.

Privacy: Ensure that the use of AI respects privacy, does not disclose sensitive information, and complies with applicable data protection laws.

Bias Awareness: Be critical of potential biases in AI outputs and address them in your methodology.

Transparency: When presenting your findings, be transparent about the extent of AI usage and its impact on your results

Submission Guidelines:

Submit your report via the course submission portal (Turnitin assignment) by **May 15th, 2026 (Midnight, Madrid Time)**. Ensure that all deliverables meet the stipulated requirements and are submitted on time. Late submissions will not be allowed. Upload your video to the following [shared folder](#).

Evaluation Criteria:

Your project will be evaluated based on the following:

- Depth and accuracy of the theoretical analysis.
- Creativity and practicality of the applied solution.
- Clarity and professionalism of the report and video presentation.
- Adherence to ethical guidelines for AI usage.
- Team collaboration.

Grading Scheme for Multivariable Optimization Group Project

Report (60% of total grade)

- **Introduction and Literature Review (15%):**
 - Clarity of the problem statement and objectives.
 - Relevance and thoroughness of the literature review.
- **Theoretical Analysis (20%):**
 - Accuracy and complexity of the mathematical models.
 - Appropriate use of optimization and constrained optimization techniques.
 - Depth of understanding of the multivariable functions involved.
- **Application and Results (15%):**
 - Effectiveness of applying the theoretical model to a practical scenario.
 - Soundness of the methodology in simulations or real-world data analysis.
 - Quality and practicality of the proposed solutions.
- **Discussion and Conclusion (10%):**
 - Critical evaluation of the solution's impact, scalability, and limitations.
 - Insightful discussion of potential real-world applications.
 - Conclusive remarks that summarize the findings and suggest future work.

Video Presentation (40% of total grade)

- **Content (20%):**
 - Clear and concise explanation of the project's objectives, methodology, and results.
 - Effective communication of the project's significance.
- **Delivery (10%):**
 - Professionalism and clarity of speech.
 - Engagement with the audience.
- **Visuals and Production Quality (5%):**

- Quality of visuals and adherence to the time constraint.
- Overall production value, including editing and audio clarity.